

EXPERIMENT NO. 6

Aim: Use string functions and regex for email validation, data extraction, and text analysis.

Requirements: A computer with a web browser, Visual Studio Code (or any text editor), HTML, CSS, and JavaScript.

Theory:

A **string** in JavaScript is a sequence of characters used to represent and manipulate textual data. Strings are commonly used for handling names, messages, email addresses, passwords, and other text-based information. JavaScript provides several built-in functions to perform different operations on strings.

Some commonly used string functions are:

- **toUpperCase()** – Converts all characters of a string into uppercase.
- **toLowerCase()** – Converts all characters of a string into lowercase.
- **substring()** – Extracts a specified part of a string.
- **indexOf()** – Returns the position of a specified character or word.
- **split()** – Divides a string into an array based on a specified separator.
- **replace()** – Replaces a specified word or character with another value.
- **trim()** – Removes extra spaces from the beginning and end of a string.

A Regular Expression (Regex) is a sequence of characters that defines a search pattern. It is mainly used to search, match, validate, and extract specific patterns from text. In JavaScript, regular expressions can be used with methods such as `test()`, `match()`, `search()`, and `replace()`.

For email validation, Regex checks whether the entered email follows a proper format containing a username, `@` symbol, domain name, and extension. For example:

student@gmail.com

For data extraction, Regex can be used to extract numbers, email addresses, dates, or other specific patterns from a given text.

For text analysis, string functions and Regex can be used to count the number of words, vowels, digits, or occurrences of specific characters in a text.

Thus, string functions and Regular Expressions provide an efficient way to process, analyze, validate, and manipulate textual data in JavaScript.

Algorithm:

1. Start the program.

2. Declare a string containing text data and an email address.
3. Define a Regular Expression pattern for validating the email format.
4. Use the test() method to check whether the entered email matches the pattern.
5. Display “Valid Email” if the pattern matches; otherwise, display “Invalid Email”.
6. Use the match() method with a Regex pattern to extract numerical or required data from the given text.
7. Use trim() and split() functions to separate the text into words and count the total number of words.
8. Use a Regex pattern to find all vowels and calculate their count.
9. Apply other string functions such as toUpperCase() and toLowerCase() for text manipulation.
10. Display all the obtained results.
11. Stop the program.

Conclusion:

Thus, the experiment on String Functions and Regular Expressions in JavaScript was successfully performed. Various string methods were studied to manipulate and process textual data. Regular Expressions were used to validate an email address and extract specific data such as numbers from a given text. Text analysis was also performed by counting words and vowels. This experiment demonstrates the importance of string handling and Regex in developing applications that require data validation, searching, extraction, and text processing.

String Operations

Enter a string:

Process

Original String: Gitika

Reversed String: akitiG

Vowels: i, i, a

Number of Vowels: 3

3. Text Analysis

hello world abcd

Analyze Text

Character Count: 17
Word Count: 3
Vowel Count: 4
Number Count: 0
Sentence Count: 1
Contains "JavaScript": No
Uppercase:
HELLO WORLD ABCD
Lowercase:
hello world abcd

✓ Valid Email Address

2. Email/Data Extraction

gitikamakheja@gmail.com, 76589035556,k3ejid

Extract Data

Emails:
gitikamakheja@gmail.com
Phone Numbers:
6589035556
URLs:
No URLs found.

3. Text Analysis

String Functions & Regular Expressions

1. Email Validation

gitikamakheja@gmail.com

Validate Email

✓ Valid Email Address

2. Email/Data Extraction

5679825261

Extract Data